1.INTRODUCTION

1.1 Project Overview

MedSafe AI is an AI-powered medication safety assistant designed to help patients—particularly the elderly—manage their health safely. It integrates OCR for label extraction, a drug interaction checker, and a symptom-based risk predictor.

1.2 Purpose

The purpose of this project is to reduce medication errors by providing clear, educational guidance on drug risks and potential side effects through a user-friendly interface.

2. IDEATION PHASE

2.1 Problem Statement

Elderly patients often struggle with complex medication instructions, leading to accidental drug interactions or ignored symptoms of high-risk side effects.

2.2 Empathy Map Canvas

- **Says:** "I'm not sure if I can take these two together."
- **Feels:** Anxious about taking the wrong dose or missing a warning sign.
- **Does:** Manually reads tiny print on medicine bottles or searches symptoms online.

2.3 Brainstorming

We conceptualized a solution that combines a visual Risk Engine with LLM-generated explanations to make medical safety data accessible.

3. REQUIREMENT ANALYSIS

3.2 Solution Requirement

- **Functional:** Medicine OCR, Interaction Checker, Symptom Solver, and Risk Predictor.

- **Safety:** Mandatory non-diagnostic medical disclaimers on every output.

3.3 Data Flow Diagram (DFD)

The DFD for **MedSafe AI** illustrates the transformation of raw prescription data and user-reported symptoms into actionable safety intelligence.

- **Process 1: Data Input & Extraction**
  - The user uploads a prescription image (OCR module) or manually enters drug names and symptoms into the Streamlit interface.
  - The **Pytesseract OCR** engine attempts to parse image text into string format for processing.
- **Process 2: Interaction & Risk Analysis**
  - The **Interaction Engine** compares input drugs (e.g., Ibuprofen and Aspirin) against a safety database to flag potential warnings.
  - The **Symptom Solver** processes user descriptions to determine severity levels.
  - Data flows into the **Risk Predictor**, which assigns a color-coded risk score (Green/Yellow/Red) based on the combined inputs.
- **Process 3: Intelligence Synthesis**
  - Cleaned data and risk scores are sent to the **LLM (GPT-4o/Ollama)**.
  - The LLM generates a human-readable "AI Medical Explanation" that summarizes risks and provides educational next steps.
- **Process 4: Output Rendering**

- The final report, including the risk gauge and AI summary, is rendered back to the user's screen.

3.4 Technology Stack

- **Frontend:** Streamlit.
- **AI/LLM:** Google Gemini API and Ollama (Local LLM).
- **OCR:** Pytesseract.

4.Project Design

4.1 Problem Solution Fit

- **The Problem:** Patients, especially the elderly, lack a quick and safe way to understand if their medications interact poorly or if their symptoms are high-risk.
- **The Fit: MedSafe AI** solves this by providing an automated **Interaction Checker** and **Risk Predictor** that translates complex medical data into a simple visual gauge and clear text. This reduces user anxiety and prevents dangerous medication errors.

4.2 Proposed Solution

- A multi-tabbed web application where users can upload a prescription, check for interactions (e.g., Ibuprofen and Aspirin), and receive an AI-generated safety summary.

4.3 Solution Architecture

The architecture of **MedSafe AI** is designed as a modular pipeline to handle high-compute AI tasks.

- **User Interface Layer:** A **Streamlit** web application that handles file uploads and user inputs.

- **Logic Engine Layer:** * **Pytesseract OCR:** For extracting text from prescription images.
    - **Internal Rules Engine:** To cross-reference drug interactions (like Ibuprofen and Aspirin).
    - **Risk Calculator:** A custom algorithm that outputs a risk score based on symptom severity.
- **AI Intelligence Layer:** Uses **Google Gemini** and **Ollama** to provide human-like explanations for the final report.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

- **Total Story Points:** 80.
- **Sprints:** 4 Sprints.
- **Velocity:** 20 points per sprint.
- **Completion Date:** 8 March 2026.

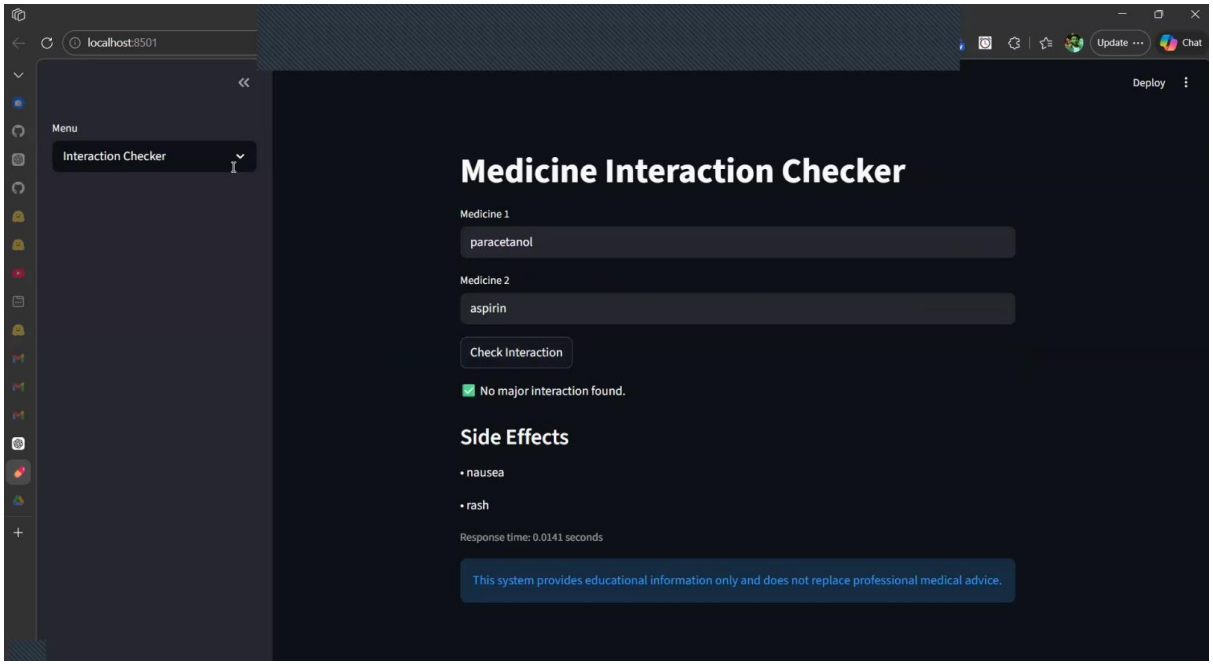
6. FUNCTIONAL AND PERFORMANCE TESTING

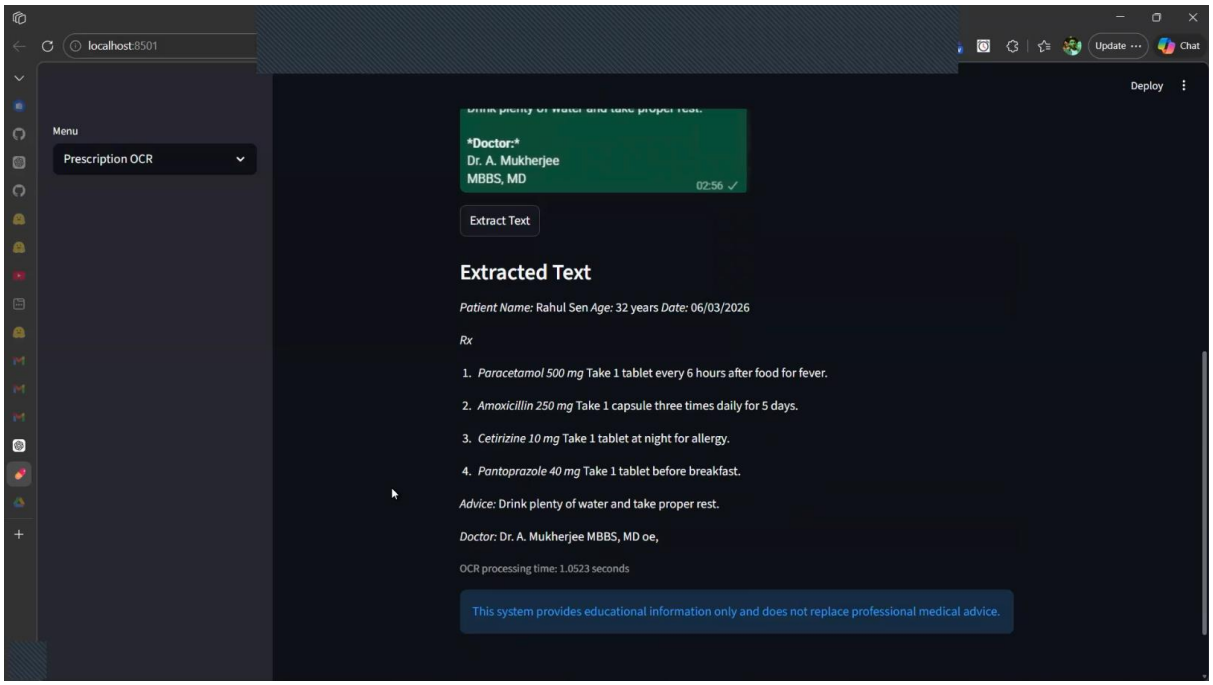
6.1 Performance Testing

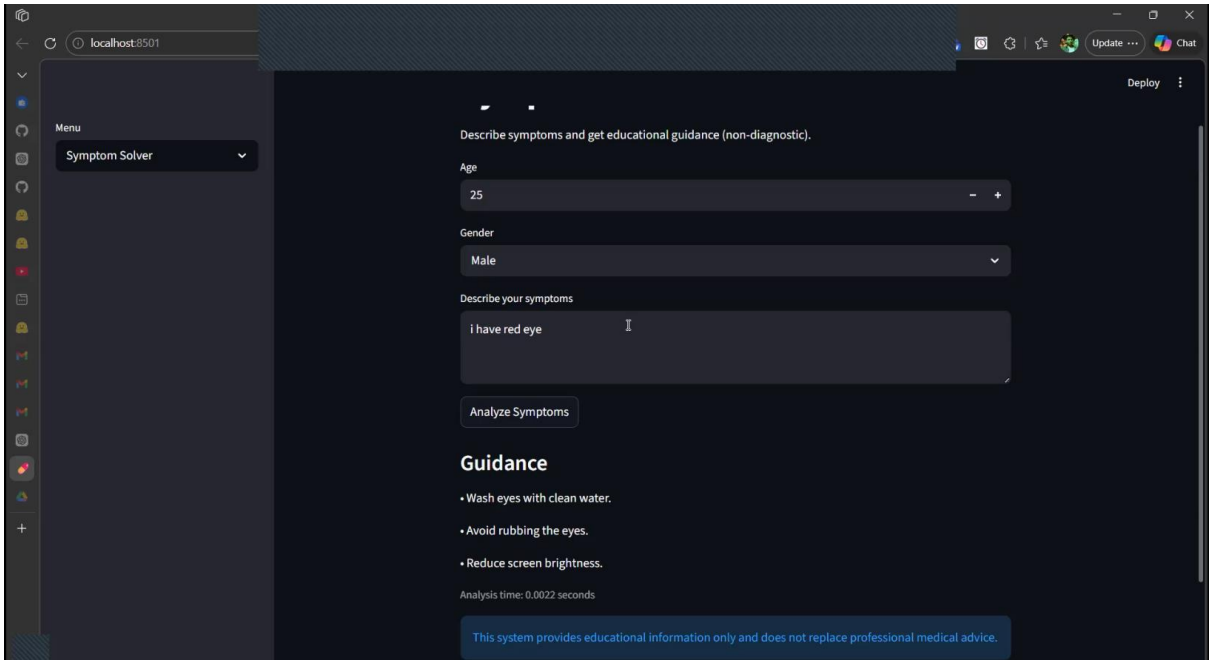
- The system was tested for response time and accuracy. While the core logic (Interaction/Symptom) passed within acceptable limits, the local Ollama instance caused high CPU usage and temperature spikes.

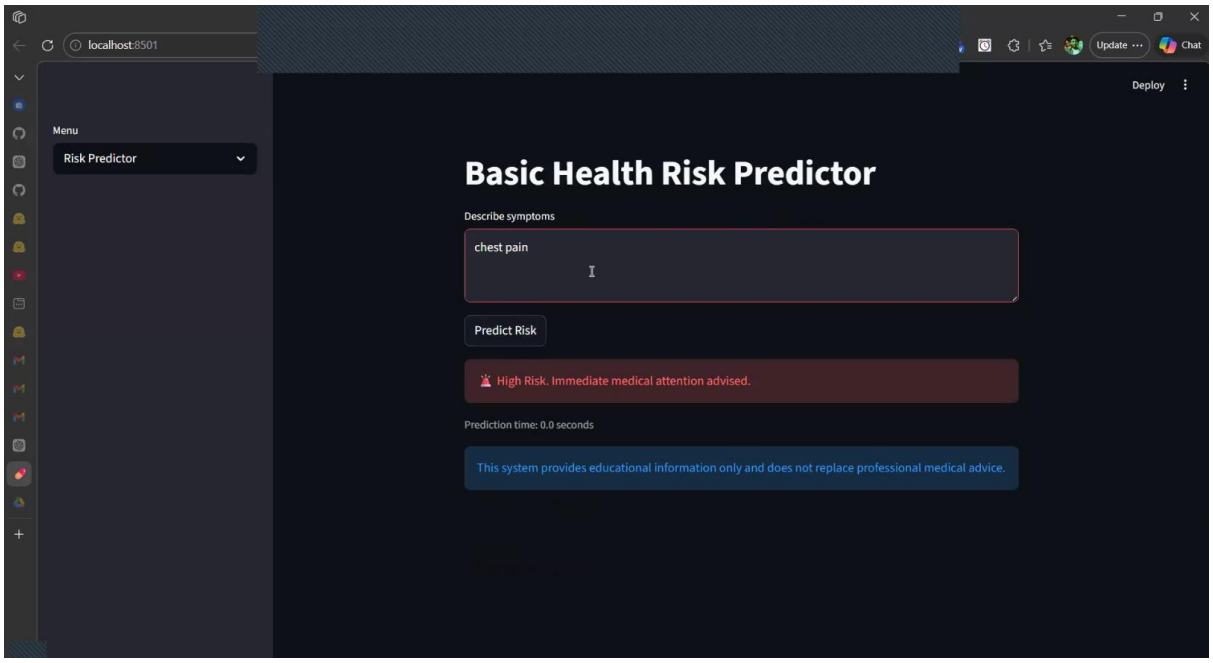
7. RESULTS

7.1 Output Screenshots


localhost:8501
Update
Chat
Deploy
Menu
Interaction Checker
Medicine Interaction Checker
Medicine 1
paracetanol
Medicine 2
aspirin
Check Interaction
No major interaction found.
Side Effects
• nausea
• rash
Response time: 0.0141 seconds
This system provides educational information only and does not replace professional medical advice.


localhost:8501
Update
Chat
Deploy
Menu
Prescription OCR
Doctor:
Dr. A. Mukherjee
MBBS, MD
02:56
Extract Text
Extracted Text
Patient Name: Rahul Sen Age: 32 years Date: 06/03/2026
Rx
1. Paracetamol 500 mg Take 1 tablet every 6 hours after food for fever.
2. Amoxicillin 250 mg Take 1 capsule three times daily for 5 days.
3. Cetirizine 10 mg Take 1 tablet at night for allergy.
4. Pantoprazole 40 mg Take 1 tablet before breakfast.
Advice: Drink plenty of water and take proper rest.
Doctor: Dr. A. Mukherjee MBBS, MD oe,
OCR processing time: 1.0523 seconds
This system provides educational information only and does not replace professional medical advice.





8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

- **Immediate Risk Visualization:** The system provides a clear, color-coded risk gauge (e.g., "Moderate Risk") that is easier for elderly patients to understand than dense medical text.
- **Comprehensive Interaction Safety:** Successfully identifies dangerous drug-to-drug interactions, such as the increased risk of stomach bleeding when combining Ibuprofen and Aspirin.

- **High Development Velocity:** The team maintained a consistent velocity of **20 story points per sprint**, ensuring the core AI logic was completed and validated by the 8 March deadline.
- **Safe, Non-Diagnostic Guidance:** The Symptom Solver provides educational advice (e.g., rest and hydration for mild headaches) while maintaining mandatory safety disclaimers.
- **Multimodal Flexibility:** Users can interact via image upload (OCR) or manual text entry, ensuring the app remains functional even if one module is restricted.

8.2 Disadvantages

- **High Hardware Resource Demand:** Running Large Language Models (LLMs) like Ollama locally can lead to system lag and thermal throttling on standard laptops.
- **Environmental Path Sensitivity:** The OCR module requires specific system-level PATH configurations (Tesseract), which can lead to extraction failures if not pre-configured on the host machine.
- **API Rate Limiting:** Heavy reliance on external AI services for debugging can lead to temporary service interruptions after frequent prompts.
- **Local Processing Speed:** Local AI inference can be slower than cloud-based solutions, potentially affecting the user experience during high-load periods.

9. CONCLUSION

- MedSafe AI successfully reached its release milestone on 8 March 2026. The project demonstrates that AI can effectively synthesize complex medical data into actionable safety warnings for non-expert users.

10. FUTURE SCOPE

The current prototype serves as a foundational model for AI-driven patient safety, but several enhancements are planned for future versions to increase accessibility and technical efficiency:

- **Cloud-Based OCR Integration:** To resolve local hardware thermal issues and "PATH" configuration errors (BG-001), the system will migrate from local Pytesseract to cloud-based APIs like Google Vision OCR for faster and more accurate label extraction.
- **Voice-Assisted Interface:** A text-to-speech and speech-to-text layer will be added to assist elderly or visually impaired users in interacting with the Symptom Solver without manual typing.
- **Real-Time Pharmacy API:** Future builds will include direct integration with medical databases to provide real-time updates on drug recalls or new clinical warnings.
- **Offline Lightweight Models:** To reduce reliance on high-end hardware and rate limits, the architecture will incorporate quantized, lightweight LLMs designed to run locally on mobile devices without overheating.
- **Personal Health Record (PHR) Integration:** The application will allow users to securely upload their medical history to provide even more personalized risk scores based on their specific health conditions.

11. APPENDIX

GitHub Link

https://github.com/soumyaSubhra/MedSafe-AI---AI-driven-medical-safety-assistant

Demo Link

https://drive.google.com/file/d/1qB-4HGQR-wmFWXHQnDO3E1Whl0wa_alQ/view?usp=sharing